THE NOVA RESEARCH CURRICULUM

# Computer Science and skills

Grade 7 · Term 2

Student edition

Outcomes: CS-7.1 (part) functions; CS-7.2 spreadsheets; RO-7.1; FL-7.1; RP-7.1; AR; PE.

## Lesson notes

- **CS-7.1 Functions (Weeks 1–4):** `def area(l, w):` then `return l * w`; parameters and return values; call a function many times. Program: a function that converts °C to °F (links to Physics).
- **CS-7.2 Spreadsheets (Weeks 5–12):** cells, rows, columns; formulas `=SUM(B2:B10)`, `=AVERAGE(B2:B10)`, `=MAX()`, `=MIN()`; copy formulas down; sort; make a bar or line chart. Use the class survey data from Research.
- **RO-7.1:** control a motor in the simulator; turn on a fan when a temperature sensor reads hot; mini project.
- **FL-7.1:** what taxes pay for (roads, schools, hospitals); sales tax on a bill; reading an electricity bill.
- **RP-7.1:** check the survey data for mistakes; make charts; draft findings.
- **AR, PE:** poster design and lettering; hockey skills and team tactics; health talk on the harm of smoking and drugs.

## Workbook pages

| Page | Task |
|---|---|
| 1 | Write and test three functions. |
| 2–3 | Spreadsheet of class marks with formulas and a chart. |
| 4 | Fan-control circuit log. |
| 5 | Bill-reading worksheet. |

## Question bank

| # | Outcome | Question | Your answer |
|---|---|---|---|
| 1 | CS-7.1 | Which keyword starts a function in Python? | func / def / function / start |
| 2 | CS-7.1 | What does `return` do? | prints text / sends a value back from the function / ends Python / makes a loop |
| 3 | CS-7.1 | Write a function that doubles a number. | _____ |
| 4 | CS-7.2 | Which formula adds cells B2 to B10? | =ADD(B2:B10) / =SUM(B2:B10) / =B2+B10 / =TOTAL(B2-B10) |

| # | Outcome | Question | Your answer |
|---|---|---|---|
| 5 | CS-7.2 | Which formula finds the mean? | =MEAN() / =MID() / =AVERAGE() / =MODE() only |
| 6 | CS-7.2 | A line chart is best for showing change over time. True or false. | True / False |
| 7 | RO-7.1 | A motor is an ... device. | input / output / storage / sensor |
| 8 | FL-7.1 | Taxes help pay for public schools and roads. True or false. | True / False |